

GROWTH MINDSET INTERVENTION: CAUSAL EFFECTS ON STUDENT SUCCESS

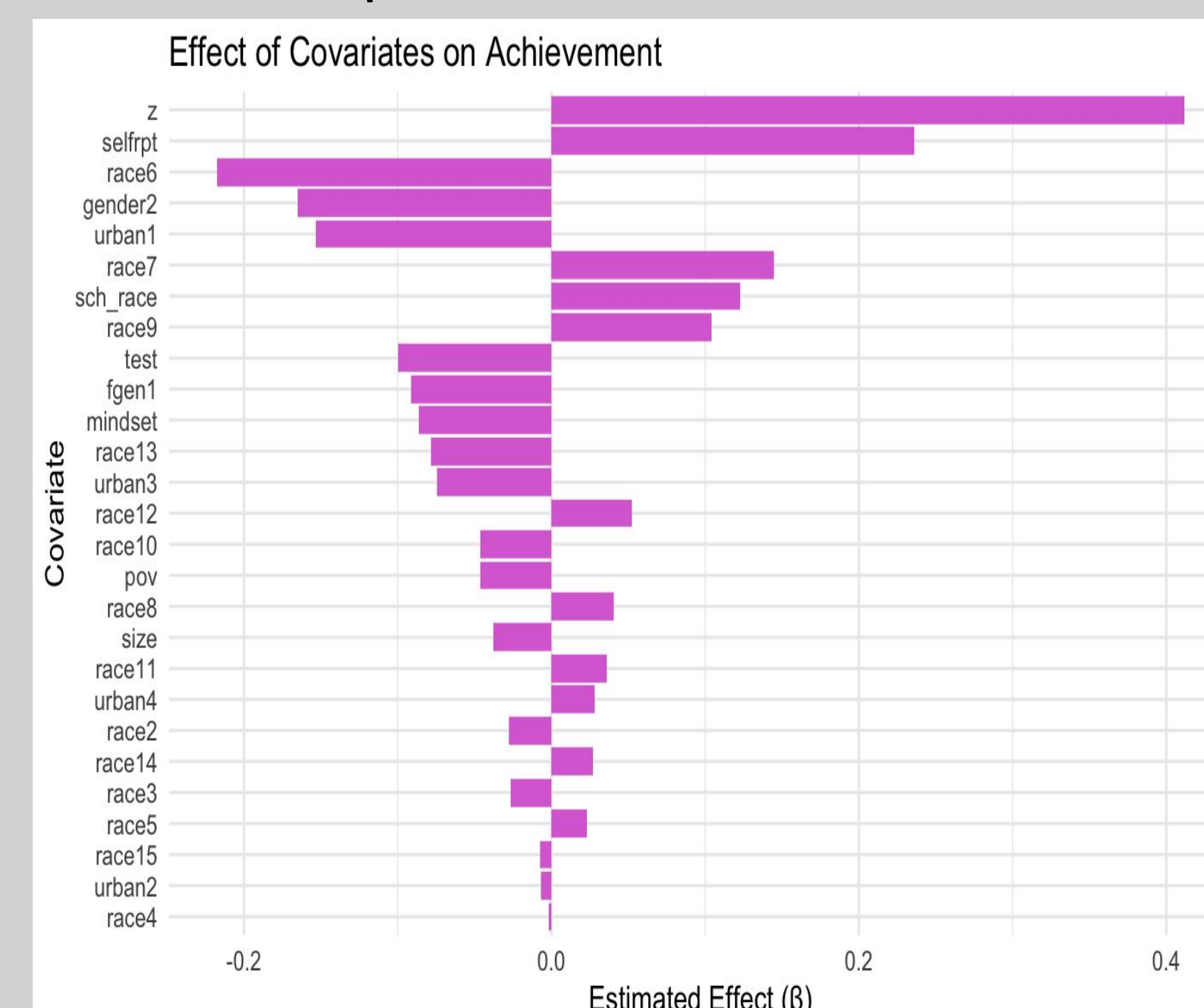
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INTRODUCTION

A growth mindset - the belief that intelligence develops over time, has been proven to improve academic success. This project studies the causal effect of the “nudge-like” intervention using a synthetic dataset which is based on the National Study of Learning Mindsets experiment. Since the data is observational and not fully randomized, causal inference methods are applied to estimate the Average Treatment Effect (ATE) of the intervention on student achievement.

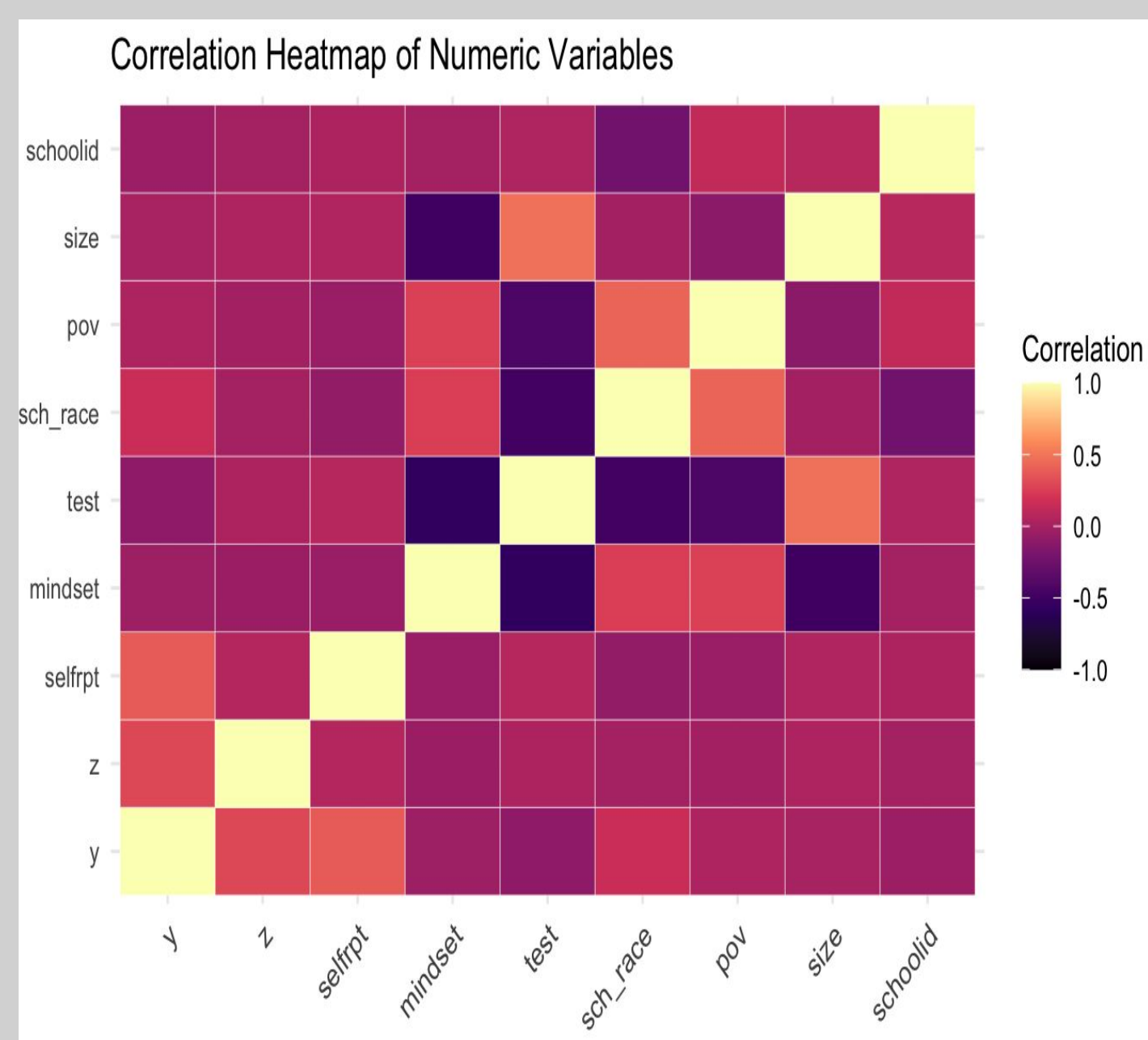
DATA EXPLORATION

Prior to implementing causal inference methods, the data was analyzed to identify key variables, explore distributions, and visualize patterns.

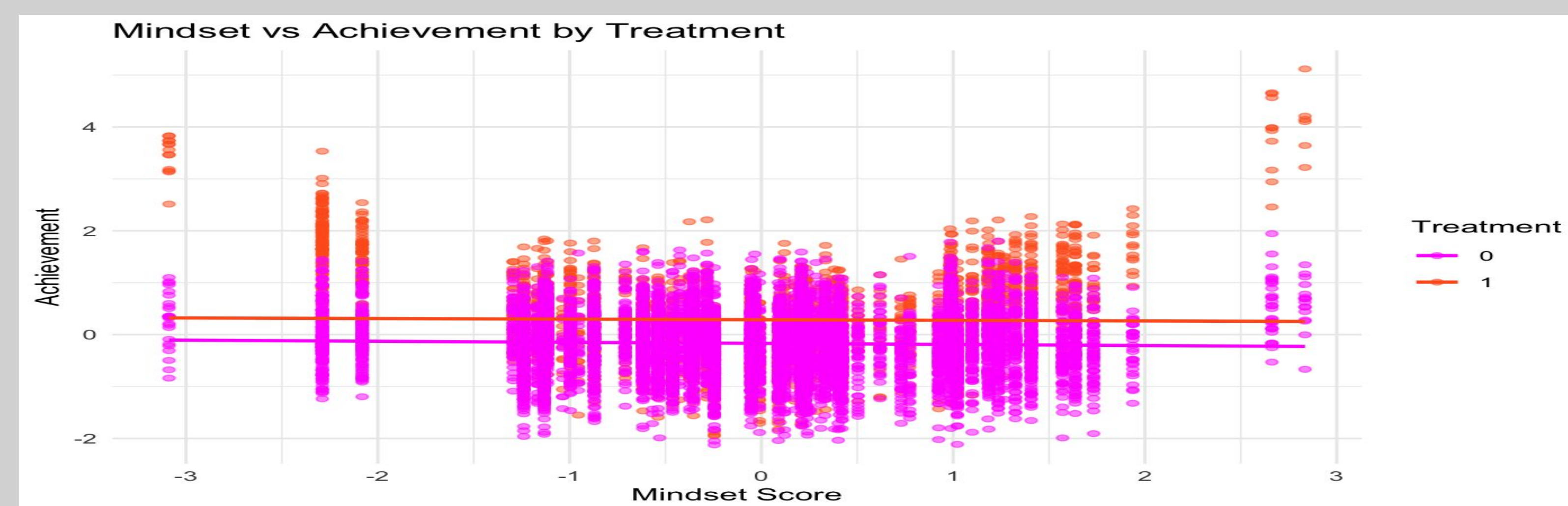


The regression coefficients suggest that both the treatment and self-confidence have strong positive effects on achievement, while certain demographic variables exhibit notable negative associations.

Strong correlations are observed between achievement, self-confidence and test performance (test). Other variables like poverty level and school race show moderate correlations with each other which may suggest possible socio-demographic confounding.



Among students who received the treatment ($z = 1$), higher mindset scores are linked to better achievement, whereas in the control group ($z = 0$), the relationship appears flat, suggesting a potential interaction between the intervention and mindset in shaping performance.



ASSUMPTIONS

To ensure that the causal inferences drawn are unbiased, the following assumptions are made.

- 1. Conditional Ignorability (Unconfoundedness)** - All important confounders must be included in the model – otherwise, bias remains.
- 2. Overlap (Positivity)** - Every student must have a positive probability of being in either the treated or control group.
- 3. Model Specification** - At least one of the models must be reasonably accurate for the ATE to be correct.

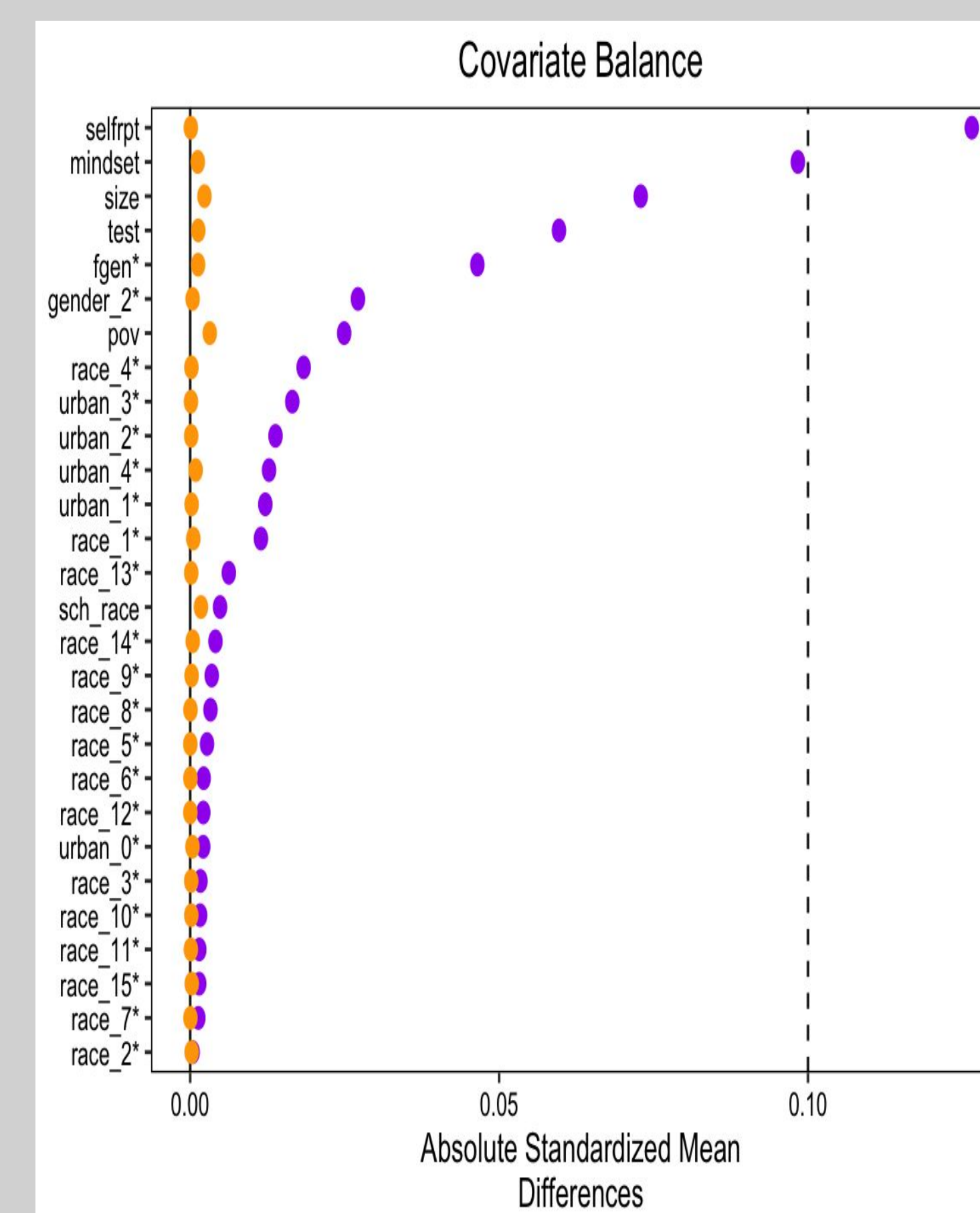
CAUSAL INFERENCE METHODS

Linear Regression

- Linear regression led to an estimated 0.41 increase in achievement, with a tight 95% confidence interval of [0.387, 0.436].
- This result is statistically significant, highlighting a clear positive effect of the growth mindset intervention.

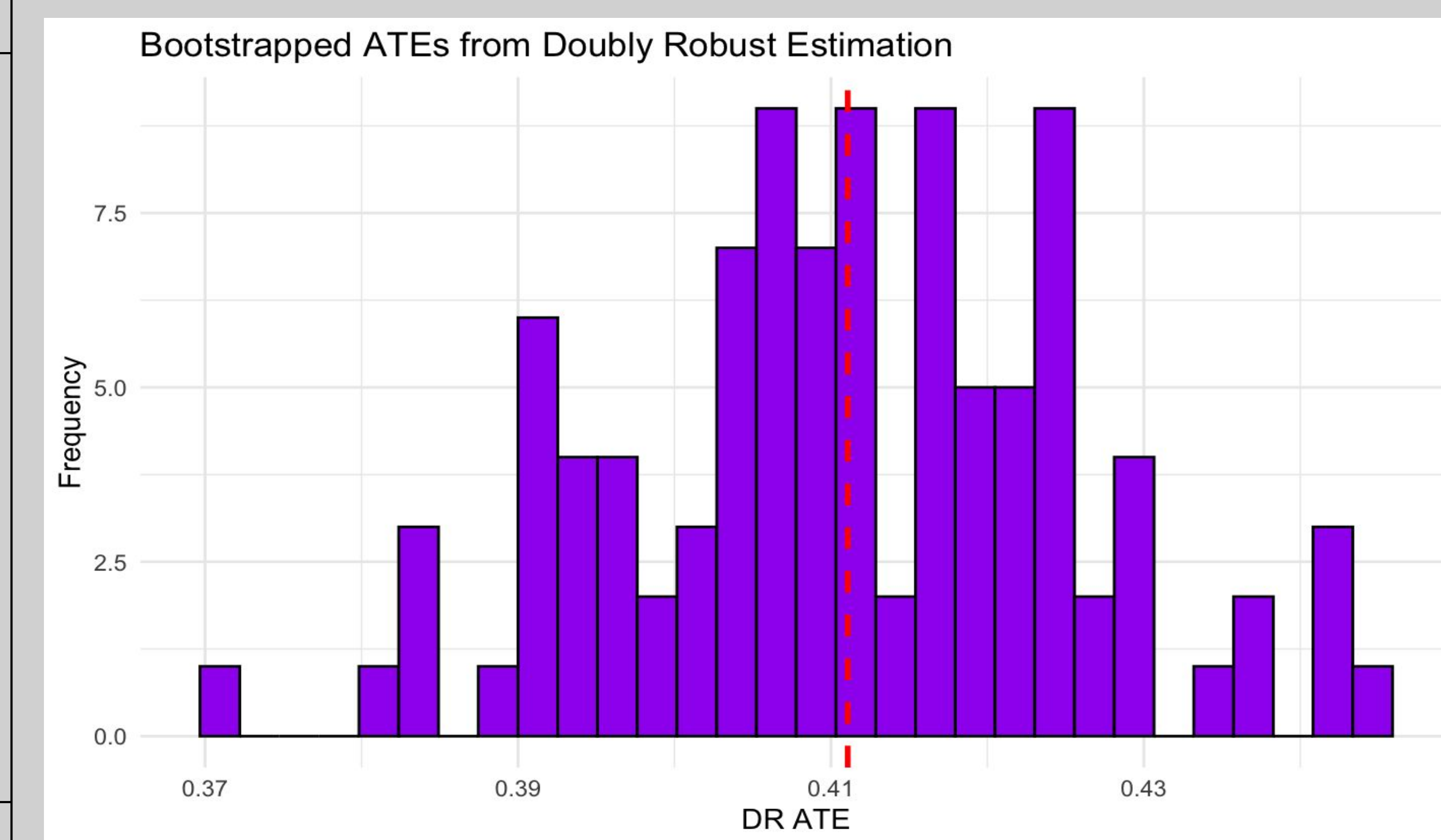
Inverse Probability of Treatment Weighting (IPTW)

- IPTW reweights the sample using estimated propensity scores to create a pseudo-population to mimic a randomized experiment and gave similar results as linear regression.
- This love plot shows that all covariates achieve excellent balance with standardized mean differences below the 0.1 threshold.



Doubly Robust Estimation

- This method works by predicting what each student’s achievement would be with and without the treatment, then adjusts these predictions using their chance of getting the treatment to get a more accurate estimate of the treatment’s effect.
- The histogram displays the distribution of bootstrapped ATEs awhere the normal estimate centered around 0.41.



CONCLUSION

The growth mindset intervention consistently improves student achievement, with estimates from linear regression (ATE = 0.412), IPTW (ATE = 0.41), and doubly robust estimation (ATE = 0.410). Covariate balance plots and supporting visualizations show that weighting improved comparability between groups. The results indicate a positive and statistically significant impact after adjusting for observed confounders.

LIMITATIONS

- The analysis assumes no unmeasured confounding, which cannot be empirically verified in observational data.
- Additionally, generalizability may be limited if the sample is not representative of broader populations.
- Estimates may be sensitive to model misspecification if either model is poorly fitted.